

Lesson 24 Inclusion-Exclusion Principle

1. Prove or disprove:
 - a) If A, B , and C are pairwise disjoint (each pair is disjoint), then
$$|A \cup B \cup C| = |A| + |B| + |C|$$
 - b) If $A \cap B \cap C = \emptyset$, then $|A \cup B \cup C| = |A| + |B| + |C|$.
 - c) If $|A \cup B \cup C| = |A| + |B| + |C|$, then A, B , and C are pairwise disjoint.
 - d) If $|A \cup B \cup C| = |A| + |B| + |C|$, then $A \cap B \cap C = \emptyset$.
2. How many integers between 1 and 100 (inclusive) are not a multiple of 2 or 5?
3. How many integers between 1 and 1000 (inclusive) are not a multiple of 2, 3, or 5?